CA01

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1 Definitions

1.1 Problem Modelling

The task is a supervised multiclass classification problem predicting diagnosis (Depression, CI, or Both) from demographic and clinical features. Regression and clustering are unsuitable because the target variable is categorical and already labeled. The dataset contained 670 observations and 19 variables.

1.2 Data Analysis and Feature Selection

Initial exploration confirmed no missing values and no duplicate observations. Variables were classified by statistical meaning as per Figure 1. This guided encoding and scaling decisions. The id column was removed as a unique identifier, and country was removed as a constant feature. A value of 99 in *depression_phq9_score* lies outside the valid PHQ-9 clinical range 0 - 27 and was removed, leaving 669 observations. Class distribution analysis revealed moderate imbalance, with the Both class containing fewer observations, motivating the use of stratified sampling.

Variable Type	Variables
Nominal	diagnosis
Ordinal	social activity level, exercise frequency, work status
Binary	gender, meditation or mindfulness
Discrete Numerical	depression phq9 score, pem duration hours
Continuous Numerical	age, stress level, sleep quality index, fatigue severity scale score
Identifier	id *(removed)*
Constant Feature	country *(removed)*

Figure 1: Preprocessing variables

Figure 1 summarises the final variable classification used for preprocessing.

Visualisations explored the dataset. Histograms and violin plots examined numerical distributions, boxplots flagged potential outliers, and scatterplots revealed relationships with diagnosis. Fatigue tended to be higher in the CI and Both groups, while stress was distributed across all classes. Observations outside boxplot whiskers were inspected and retained as they remained within clinically valid ranges.

Class distribution analysis revealed moderate imbalance, with the *Both* class containing fewer observations (Figure 2). This motivated the use of stratified sampling in the experimental protocol.

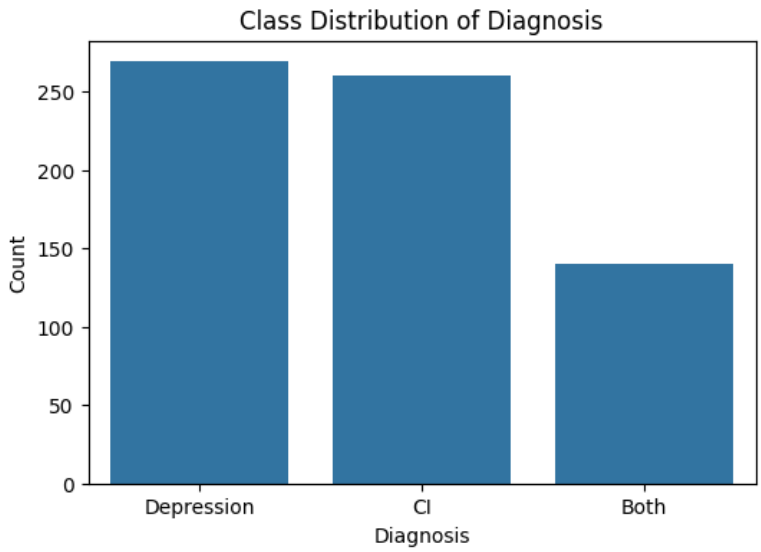

Figure 2: Diagnosis Class Distribution

These exploratory analyses provided insight into the structure of the dataset and informed subsequent feature selection and modelling decisions.

1.3 Correlation and Redundancy Analysis

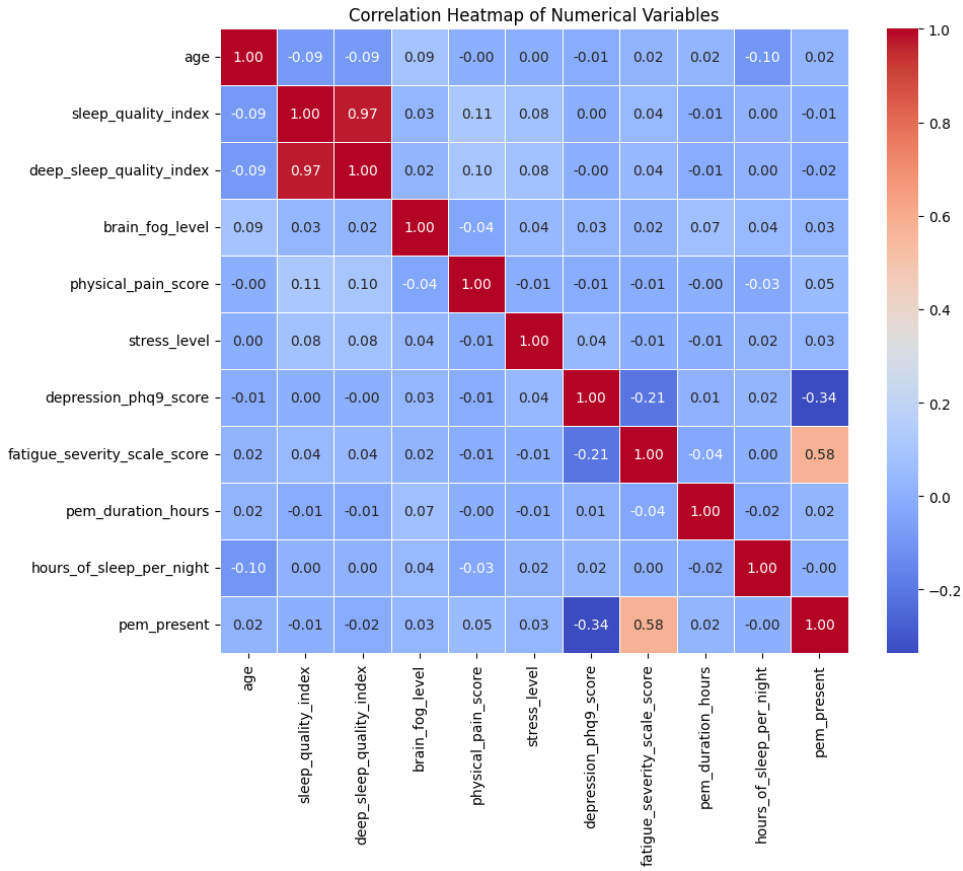

Figure 3: Correlation Heatmap

A Pearson correlation heatmap (Figure 3) revealed a very strong correlation ($r = 0.97$) between sleep quality index and deep sleep quality index. Point plots and one-way ANOVA tests were used to assess differences across diagnosis groups; neither variable showed statistically significant differences. However, *sleep_quality_index* displayed slightly clearer group separation in the point plots and was therefore retained, while *deep_sleep_quality_index* was removed. The remaining numerical variables showed only weak to moderate correlations.

1.4 Categorical Association Analysis

Cramer's V was computed to assess associations between categorical predictors and diagnosis. Most predictors showed weak associations $V < 0.10$. However, pem_present showed a perfect association $V = 1.000$, indicating it deterministically separates the diagnosis categories and therefore represents target leakage. It was removed prior to model training. After removal, all remaining categorical predictors showed negligible inter-associations (Figure 4).

Variable	Cramér's V
gender	0.013
work_status	0.082
social_activity_level	0.103
exercise_frequency	0.084
meditation_or_mindfulness	0.029
pem_present	1.000

Figure 4: Categorical associations with diagnosis

The results indicate weak associations between most categorical predictors and the diagnosis variable. For example, *gender* and *meditation_or_mindfulness* show almost no relationship with diagnosis, while *work_status*, *exercise_frequency*, and *social_activity_level* exhibit only weak associations.

After removing this variable, Cramér's V values among the remaining categorical predictors were negligible ($V < 0.10$), indicating no strong associations between retained features.

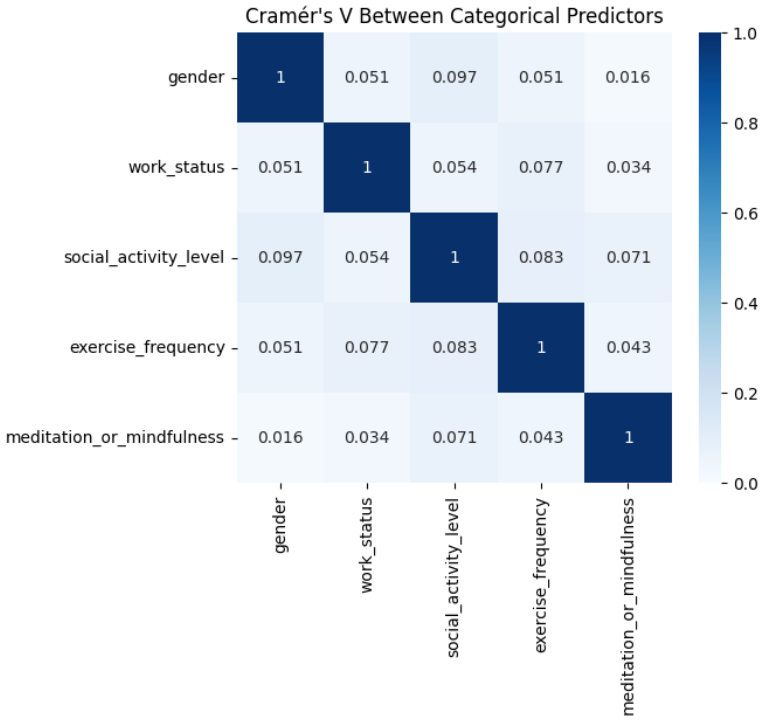

Figure 5: Categorical predictor associations (excluding *pem_present*)

1.5 Experimental Protocol

The dataset was split 80/20 with stratification. Model selection and hyperparameter tuning used stratified 5-fold CV, with macro F1-score as the primary metric. Categorical variables were encoded

(nominal: one-hot; ordinal: ordinal; binary: 0/1), and numerical features were standardised for models sensitive to feature scale (Logistic Regression, k-NN). Tree-based models used raw values. All encoding and scaling were applied within a preprocessing pipeline during cross-validation to prevent data leakage.

1.6 Selected ML Methods

Three classifiers were evaluated: Logistic Regression (linear baseline), k-Nearest Neighbours (distance-based, non-parametric), and Random Forest (nonlinear ensemble). This selection allows comparison of linear, local-similarity, and ensemble-based approaches to understand which class of decision boundary best captures the diagnostic structure.

1.7 Hyperparameter Search

Hyperparameter tuning was conducted using GridSearchCV with stratified 5-fold cross-validation, optimising the Macro F1-score. Stratification preserves class distribution across folds, which is important due to moderate class imbalance. The search space included C for Logistic Regression, k for k-Nearest Neighbours, and *the number of trees* and *maximum depth* for Random Forest. The best configuration for each model was selected based on the highest cross-validated Macro F1-score

Model	Hyperparameter Grid	Best Hyperparameters	CV Macro F1
Logistic Regression	$C \in \{0.01, 0.1, 1, 10, 100\}$	$C = 10$	0.774
k-Nearest Neighbours	$k \in \{3, 5, 7, 9, 11\}$	$k = 11$	0.548
Random Forest	$n\ estimators \in \{200, 500\}$, $max\ depth \in \{None, 10, 20\}$	$n\ estimators = 500$, $max\ depth = 10$	0.947

Figure 6: Hyperparameter tuning with 5-fold CV

2 Results

2.1 Decisions Based on Data Analysis

Decisions based on the exploratory, correlation, and categorical association analyses presented in Sections 1.2–1.4 guided data preprocessing and feature selection. Logistic Regression achieved a Macro F1-score of 0.774 with $C = 10$, while KNN performed worse (Macro F1 = 0.548, $k = 11$). Random Forest achieved the best performance (Macro F1 = 0.947 with $n_estimators = 500$ and $max_depth = 10$), indicating that nonlinear relationships and feature interactions play an important role in predicting diagnosis.

2.2 Model Comparison

Model	Cross-Validated Macro F1 Score
Logistic Regression	0.774
k-Nearest Neighbours	0.548
Random Forest	0.947

Figure 7: Cross-validated model comparison

Logistic Regression achieved a Macro F1-score of 0.774 with $C = 10$, indicating that weaker regularisation improved performance. The KNN model achieved a lower score of 0.548 with $k = 11$, suggesting that local neighbourhood similarity alone was insufficient to separate the classes effectively. (Figure 6).

Additionally, the feature space in this dataset is heterogeneous, containing a mix of numerical and encoded categorical variables with different scales and distributions, which further distorts distance calculations. The relatively large optimal k = 11 also suggests that the decision boundaries are not well-captured by local neighbourhood structure alone.

Random Forest achieved the highest cross-validated Macro F1-score of 0.947 with $n_estimators = 500$ and $max_depth = 10$. This indicates that nonlinear relationships and feature interactions play an important role in predicting diagnosis.

The large gap of Macro F1-scores confirms that nonlinear relationships and feature interactions drive diagnostic prediction. Logistic Regression's moderate score indicates some linear structure exists, while KNN's weak performance reflects the curse of dimensionality, in heterogeneous, high-dimensional feature spaces, distance-based reasoning becomes unreliable. Random Forest was selected as the final model.

3 Final Model Evaluation

The best-performing Random Forest model ($n_estimators = 500$, $max_depth = 10$) was evaluated on the held-out 20% test set, achieving a test accuracy of 0.963 and a Macro F1-score of 0.954. The CI class was predicted perfectly (F1 = 1.000), Depression achieved F1 = 0.955 with high recall (0.981), and the Both class showed slightly lower recall (0.857), with four observations misclassified as Depression, reflecting clinical overlap between these two categories. The close agreement between the cross-validated score (0.947) and the test score (0.954) indicates the model generalises well without substantial overfitting.

Metric	Value
Test Accuracy	0.963
Test Macro F1	0.954

Figure 8: Random Forest test performance

Per-class results are summarised in the classification report and confusion matrix below

Classification Report

	precision	recall	f1-score	support
Both	0.96	0.857	0.906	28.0
CI	1.0	1.0	1.0	52.0
Depression	0.93	0.981	0.955	54.0
accuracy	0.963	0.963	0.963	0.963
macro avg	0.963	0.946	0.954	134.0
weighted avg	0.963	0.963	0.962	134.0

Figure 9: Classification report for RF

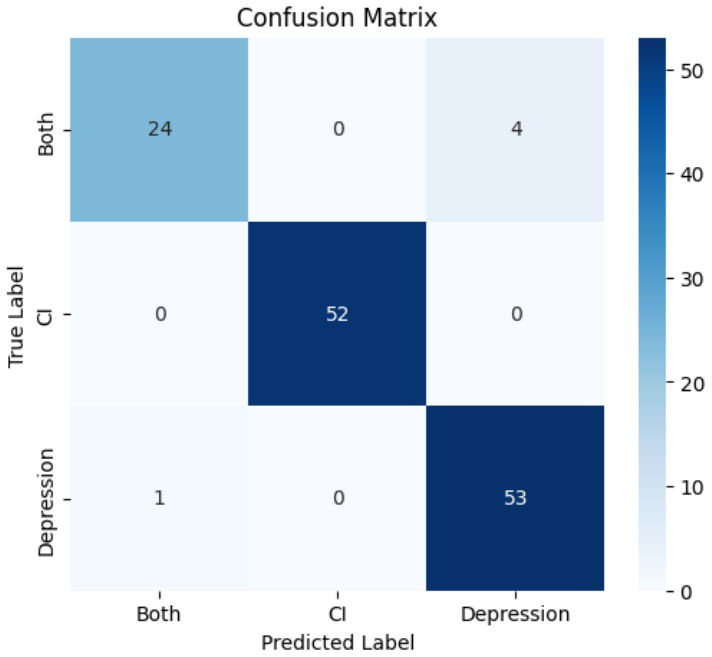

Figure 10: Confusion matrix for RF

Conclusion

This study evaluated machine learning methods for predicting diagnostic outcomes using demographic and clinical variables. Exploratory data analysis and feature screening were used to remove redundant and leakage-prone variables before modelling.

Among the evaluated methods, Random Forest achieved the best performance, indicating that nonlinear relationships and feature interactions are important for this task. The final model demonstrated strong performance on the held-out test set, suggesting good generalisation.

Future work could explore additional ensemble methods, alternative feature engineering strategies, or larger datasets to further improve predictive performance.